SQA Plan

It also should contain the schedules for the following audits proposed for the project:

- Periodic conformance audits
- Phase-end audits
- Investigative audits (and criteria)
- Delivery audits

Chapter 3: Reviews

Reference

This chapter refers from the book:

Mastering Software Quality Assurance: Best Practices, Tools and Techniques for Software Developers

Introduction to Software Testing

Outline

3.1. Review objectives

3.2. Formal review

3.3. Peer reviews

3.4. Implementation of reviews activity in projects

3.4.1. Reviewing requirement specification

3.4.2. Reviewing design document

3.4.3. Reviewing development plan and SQA plan

3.4 Exercise

Outline

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3.4 Exercise

Review

- The **design document** is key.
- It is checked **repeatedly** in the development process.
- Typically, reviewed many times before getting a stamp of approval to proceed with development.
- Unfortunately, we often don't find our own errors and thus we need others for reviews.
- Different stakeholders with different viewpoints are used in the review process.

Review

- A **review process** is : "a process or meeting during which a work product or set of work products is presented to project personnel, managers, users, customers, or other interested parties for comment or approval." (IEEE)

- Essential to detect / correct errors in these earlier work products because the **cost of errors downstream is very expensive**!!

- Review Choices:
  - Formal Design Reviews (FDR)
  - Peer reviews (inspections and walkthroughs)
  - Used especially in design and coding phase

Review objectives

Direct objectives – Deal with the current project

a. To detect analysis and design errors as well as subjects where corrections, changes and completions are required

b. To identify new risks likely to affect the project.

c. To locate deviations from templates, style procedures and conventions.

d. To approve the analysis or design product. Approval allows the team to continue on to the next development phase.

Indirect objectives – are more general in nature.

a. To provide an informal meeting place for exchange of professional knowledge about methods, tools and techniques.

b. To record analysis and design errors that will serve as a basis for future corrective actions. (very important)

Review objectives

- Many different kinds of reviews that apply to different objectives.
- Reviews are not randomly thrown together.
- Well-planned and orchestrated.
  - Objectives, roles, actions, participation, …. Very involved tasks.
- **Participants are expected to contribute in their area of expertise**.

- Idea behind reviews is to **discover problems <u>NOT</u>** to fix them/
- Typically fixed after review and 'offline' so to speak.

- **Very common to review design documents.**
- Thus they are **usually well-prepared** initially prior to review.

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3.4 Exercise

Formal Design Reviews

DPR –	Development Plan Review
SRSR –	Software Requirement Specification Review
PDR –	*Preliminary Design Review*
DDR –	Detailed Design Review
DBDR –	Data Base Design Review
TPR –	*Test Plan Review*
STPR –	Software Test Procedure Review
VDR –	Version Description Review
OMR –	*Operator Manual Review*
SMR –	Support Manual Review
TRR –	Test Readiness Review
PRR –	Product Release Review
IPR –	*Installation Plan Review*

Important to note that a design review can take place any time an analysis or design document is produced, regardless whether that document is a requirement specification or an installation document.

Formal Design Reviews

- Personal experience in:
  - Operators' Manual
  - Users' Manual

  - Program Maintenance Manual
  - Staff Users Manual

  - Installation / Conversion Plan
  - Test Plan(s)

  - Depend on size of project!

Participants in the Review

Review Leader

- Needs to be **external** to the project team.
- Must have knowledge and experience in development of the review type
- Seniority at least as high as the project leader.
- Good relationship with project leader and team

- Generally, this person really needs to know the project's material and should **NOT be the project leader**.
  - Would impact objectivity!
  - Would miss things entirely!

Participants in the Review

Review Team

- Needs to be from **senior members** of the team with other **senior members from other departments**.
  - Why?
  - Escalation?
- Team size should be 3-5 members.
- Too large, and we have too much coordination.
- This is a time for serious business – not lollygagging!

Preparations for the Design Review

- **Participants in the review** include the
  - Review leader,
  - Review team, and
  - Development team.
- **Review Leader:**
  - appoint team members;
  - establishes schedule,
  - distribute design documents, and more

Preparations for the Design Review

- **Review Team preparation**:
  - review document; list comments PRIOR to review session.
  - For large design docs, leader may assign parts to individuals;
  - Complete a checklist
- **Development Team preparations**:
  - Prepare a short presentation of the document.
  - Focus on main issues rather than describing the process.
  - This **assumes** review team has read document and is familiar with project's outlines.

The Design Review Session

- Review Leader is key person for sure!
- Start with short presentation (development team)
- Comments by members (review team)
- Comments discussed to determine required actions team must perform (review and development teams)

The Design Review Session

Decisions regarding design product – tells project's progress.

- Full approval - Continue on to next phase (may have minor corrections)
- Partial approval
  - Continue to next phase for some parts of project; major action items needed for remainder of project.
  - Continuation of these parts granted only after satisfactory completion of action items
  - Granted by review team member assigned to review completed actions or by the full review team or special review team or by some other forum
- Denial of Approval – demands a repeat of Design Review
  - Project has major defects, critical defects

The Design Review Report

- This is the Review Leader's **primary responsibility** following the review.
- Important to perform corrections early and minimize delays to project schedule.
- Report contains:
  - Summary of review discussions
  - Decision about project continuation
  - Full list of action items – corrections, changes, additions the project team must perform. For each item, anticipated completion date and responsible person is listed.
  - Name of review team member assigned for follow up.

Follow up Process

- The ‘follow-up person’ may be the review team leader him/herself
- Required to verify each action item fixed as condition for allowing project to continue to next phase.
- Follow up must be fully documented to enable clarification of the corrections in the future, if any.

Oftentimes Problems

- Sometimes entire parts of DR are often worthless due to
  - **inadequately prepared review team,** or
  - **intentional evasion** of a thorough review.

- Tipoff:
  - **Very short report** – limited to documented approval of the design and listing few, if any, defects
  - Short report approving continuation to next project phase in full - listing several minor defects **but no action items**.
  - A report listing several action items of varied severity **but no indication of follow-up** (no correction schedule, no documented activities, …)

Guidelines for Design Review

Design Review Infrastructure

- Develop checklists for common types of design documents.
- Train senior professionals to serve as a reservoir for Design Review teams.
- Periodically analyze past Design Review effectiveness.
- Schedule the Design Reviews as part of the project plan

The Design Review Team

- Review teams size should be limited, with 3–5 members being the optimum.

Guidelines for Design Review

The Design Review Session

- **Discuss professional issues** in a constructive way refraining from personalizing the issues.
- **Keep** to the review **agenda**.
- **Focus on detection of defects** by verifying and validating the participants' comments.
  - Refrain from discussing possible solutions.
- **If disagreement about an error** - end the debate by noting the issue and shifting its discussion to another forum.
- **Properly document discussed comments**, and the results of their verification and validation.
- Review session should **not exceed two hours**.

Guidelines for Design Review

Post-Review Activities

- Prepare the review report, including the action items
- Establish follow-up to ensure the satisfactory performance of all the list of action items

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3.4 Excercise

Peer Reviews

- Will discuss
  - 1. inspections and
  - 2. walkthroughs.
- Difference between formal design reviews and peer reviews is really in both their participants and authority.
- DRs: most participants hold superior positions to the project leaders and customer reps;
- Peer reviews, we have equals
  - members of his/her department and other units.

Peer Reviews

- Other major difference is
  - degree of authority and
  - objective of each review method.

- FDRs: authorized to approved design doc
  - work can now continue in project.
- Not granted in peer reviews
  - main objectives lie in
    - detecting errors and
    - deviations from standards.

Participants of Peer Reviews

Inspection

- Review leader
- The author
- Specialized professionals:
  - Designer
  - Coder or implementer
  - Tester

Walkthrough

- Review leader
- The author
- Specialized professionals:
  - Standards enforcer
  - Maintenance expert
  - User representative

Peer Reviews:
Inspections / Walk-Throughs

- Walkthroughs and inspection differ in formality –
- Inspections emphasize the **objective of corrective action; more formal**
- Walkthroughs limited to **comments** on document reviewed.

- Inspections also look to **improve methods** as well.
- Inspections are considered **to contribute more** to general level of SQA.

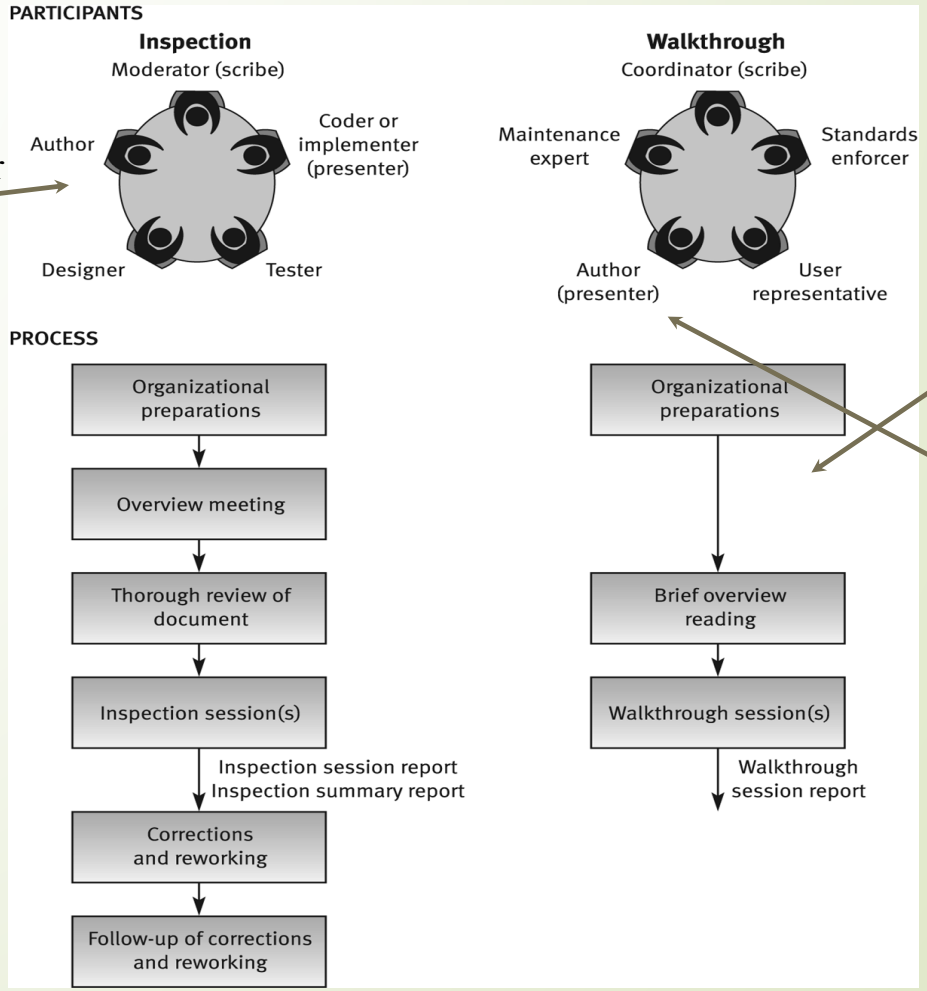
Peer Reviews: Inspections

Inspections usually based on a comprehensive infrastructure:

- Development of **inspection checklists** for each type of design document as well as coding languages, which are periodically updated.
- Development of **typical defect type frequency tables**, based on past findings to direct inspectors to potential 'defect concentration areas.'
- **Training of competent professionals** in inspection process issues – making it possible for them to serve as inspection leaders (moderators) or inspection team members
- Periodic **analysis of the effectiveness of past inspections** to improve the inspection methodology
- **Introduction of scheduled inspections** into project activity plan and allocation of the required resources, **including resources** for corrections

Inspection vs. Walkthrough

Note: author is **not** the presenter



Much less formal…

Author **is** the presenter

Focus on Peer Reviews:

- So for these two peer review methods, we will look at:
  - Participants of peer reviews
  - Preparation for peer reviews (some major differences)
  - The peer review session (presenters and emphases are different)
  - Post peer-review activities (differ considerably)
  - Peer review efficiency (arguable)
- The principles that we talk about can apply to both **design** peer reviews and **code** peer reviews.

Participants of Peer Reviews

- Optimally, 3-4 participants
- Should be **peers** of the software system designer-author.
  - This allows for free discussion without any intimidation.
- Need a good blend of individual types: a review leader, the author, and specialized professionals as needed for the focus of the review.
- **Review Leader**
  - Moderator in inspections; Coordinator in walkthroughs.
  - Must be well-versed in project development and current technologies
  - Have good relationships with author and development team
  - Come from **outside** the project team
  - History of **proven experience** in coordination / leadership settings like this.
  - For inspections, **training** as a moderator is required.

Participants of Peer Reviews

- **Specialized Professionals – note these are experienced folks.**
- These differ by review type: **inspections** / walkthroughs
- **<u>Inspections</u>**:
  - **A designer** – generally the **systems analyst** responsible for analysis and design of software system reviewed
  - **A coder or implementer** – one who is **thoroughly acquainted** with coding tasks, preferably the leader of the designated coding team.
    - Able to detect defects leading to coding errors and other software implementation issues.
  - **A tester** – experienced professional, **preferably leader of assigned testing team** who focuses on identification of design errors usually detected during the testing phase.

Participants of Peer Reviews

Walk Throughs:

- **A standards enforcer** – team member specialized in development standards and procedures;
  - locate deviations from these standards and procedures.
  - These problems substantially affect the team's long-term effectiveness for both development and follow-on maintenance.
- **A maintenance expert** – focus on maintainability / testability issues to detect design defects that may hinder bug correction and impact performance of future changes.
- **A maintenance expert** - Focuses also on documentation (completeness / correctness) vital for maintenance activity.
- **A user representation** – need an internal user (if customer is in the unit) or an external representative - review's validity due to his/her point of view as user-customer rather than the designer-supplier.

Participants of Peer Reviews

Team Assignments

- **Presenter**:
  - For **inspections:**
    - The presenter of document; chosen by the moderator; should **not be** document's author
    - Sometimes the software **coder serves as presenter** due to the familiarity of the design logic and its implications for coding.
  - For **walk-throughs**:
    - Author most familiar with the document should be chosen to present it to the group.
    - Some argue that a neutral person should be used..
- **Scribe**:
  - Team leader will often serve as the scribe and record noted defects to be corrected.

Preparation for a Peer Review Session

- Peer Review **Leader's** Preparation for the session:
- **For Design Document**:
  - Select the most difficult / complex sections; sections prone to defects.
  - The most critical sections / where defect can cause severe damage
- **Select team members**
  - Limit review session to two hours – absolutely.
  - Schedule up to two sessions a day if review tasks is sizable.
  - Schedule right after the document is ready for inspection. Don't wait….
- **Distribute the document** to the team members prior to the review session.

Preparation for a Peer Review Session

- **Peer review team's** preparations for review session:
- **For inspections:** team members, preparation is quite thorough;
  - For **walkthrough** brief.
  - **Inspection**: participants **must read document** and **list comments** before inspection begins.
    - In **overview meeting**, the author provides inspection team members with the necessary background for reviewing chosen document, project in general, logic, processes, outputs, inputs, interfaces.
    - **Tool** for inspector's review: **checklist** for specific documents.
- **For walkthroughs:** team briefly reads materials for general overview of project
  - Generally they lack detailed knowledge and its substantive area.
  - In most cases, team participants not required to prepare advance comments.

The Peer Review Session

- Procedurally, presenter reads document section and may add an explanation.
- Participants may offer comments on doc or on comments
- **Restrict discussion** to identification of errors – **no solutions**.

- **Presenter** in walkthrough provides an overview
- **Walkthrough Scribe** records each error (location, description, type) – incorrect, missing, etc.

- **Inspection scribe will add estimated severity of each defect, a factor to be used in the statistical analysis of defects found and for the foundation of preventive / corrective actions**.

The Peer Review Session

Session Documentation

- **For inspections** – much more comprehensive
  - Inspection Session Findings Report – produced by scribe
  - Inspection Session Summary Report – compiled by inspection leader after session or series of sessions dealing with the same document
    - Report summarizes inspection findings and resources invested int eh inspections…
    - Report serves as inputs for analysis aimed at inspection process improvement and corrective actions that go beyond the specific document or project.
- **For walkthroughs** – copies of the error documentation should be provided to the development team and the session participants.

The Post Review Session

- Here is the most fundamental differentiating element between the two peer review methods.

- **Inspection**:
  - **Does not end** with a review session or distribution of reports
  - Post inspection activities are conducted to attest to:
    - Prompt, effective correction / reworking of all erorr
    - Transmission of the inspection reports to controlling authority for analysis

The Efficiency of Peer Reviews

- These activities are under constant debate.
- Some of the **more common metrics** applied to estimate the effectiveness of peer reviews, suggested by literature:
  - **Peer review detection efficiency** (average hours worked per defect detected)
  - **Peer review defect detection density** (average number of defects detected per page of the design document)
  - **Internal peer review effectiveness** (percentage of defects detected by peer review as a percentage of total defects detected by the developer

The Efficiency of Peer Reviews

- (Not a lot of data on findings) An interesting study undertaken by Cusumano reports results of a study on the effectiveness of design review, code inspection, and testing at Fujitsu from 1977 to 1982.
- Findings are still of interest, **as data shows substantial improvement in software quality associated with an increased share of code inspection and design reviews and a reduced share of software testing.**
- Software quality measured here by the number of defects per 1000 lines of maintained code, detected by the users during the first six months of regular software system use.
- The results only refer to the inspection method; one **guesses** a similar result would apply to walkthrough methods.

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3.4 Excercise

Review Checklist for Software Project Planning

- Is the software scope unambiguously defined and bounded?
- Is terminology clear?
- Are resources adequate for the scope?
- Are resources readily available?
- Are tasks properly defined and sequenced?
- Is the basis for cost estimation reasonable? Has it been developed using two different sources?
- Have historical productivity and quality data been used?
- Have differences in estimates been reconciled?
- Are pre-established budgets and deadlines realistic?
- Is the schedule consistent?

Review Checklist for Software Requirements Analysis

- Is the information domain analysis complete, consistent, and accurate?
- Is problem partitioning complete?
- Are external and internal interfaces properly defined?
- Are all requirements traceable to the system level?
- Is prototyping conducted for the customer?
- Is performance achievable with constraints imposed by other system elements?
- Are requirements consistent with schedule, resources, and budget?
- Are validation criteria complete?

Review Checklist for Software Design (Preliminary Design Review)

- Are software requirements reflected in the software architecture?
- Is effective modularity achieved? Are modules functionally independent?
- Is program architecture factored?
- Are interfaces defined for modules and external system elements?
- Is data structure consistent with software requirements?
- Has maintainability been considered?

Review Checklist for Software Design (Design Walkthrough)

- Does the algorithm accomplish the desired function?
- Is the algorithm logically correct?
- Is the interface consistent with architectural design?
- Is logical complexity reasonable?
- Have error handling and "antibugging" been specified?
- Is local data structure properly defined?
- Are structured programming constructs used throughout?
- Is design detail amenable to the implementation language?
- Which are used: operating system or language dependent features?
- Is compound or inverse logic used?
- Has maintainability been considered?

Review Checklist for Coding

- Is the design properly translated into code? (The results of the procedural design should be available at this review)
- Are there misspellings or typos?
- Has proper use of language conventions been made?
- Is there compliance with coding standards for language style, comments, module prologue?
- Are incorrect or ambiguous comments present?
- Are typing and data declaration proper?
- Are physical constraints correct?
- Have all items on the design walkthrough checklist been reapplied (as required)?

Review Checklist for Software Testing (Test Plan)

- Have major test phases been properly identified and sequenced?
- Has traceability to validation criteria/requirements been established as part of software requirements analysis?
- Are major functions demonstrated early?
- Is the test plan consistent with the overall project plan?
- Has a test schedule been explicitly defined?
- Are test resources and tools identified and available?
- Has a test recordkeeping mechanism been established?
- Have test drivers and stubs been identified, and has work to develop them been scheduled?
- Has stress testing for software been specified?

Review Checklist for Software Testing (Test Procedure)

- Have both white and black box tests been specified?
- Have all independent logic paths been tested?
- Have test cases been identified and listed with expected results?
- Is error handling to be tested?
- Are boundary values to be tested?
- Are timing and performance to be tested?
- Has acceptable variation from expected results been specified?

Review Checklist for Maintenance

- Have side effects associated with change been considered?
- Has the request for change been documented, evaluated, and approved?
- Has the change, once made, been documented and reported to interested parties?
- Have appropriate FTRs been conducted?
- Has a final acceptance review been conducted to assure that all software has been properly updated, tested, and replaced?

Formal Technical Review (FTR)

- Software quality assurance activity that is performed by software engineering practitioners
  - Uncover errors in function, logic, or implementation for any representation of the software
  - Verify that the software under review meets its requirements
  - Assure that the software has been represented according to predefined standards
  - Achieve software that is developed in a uniform manner
  - Make projects more manageable
- FTR is actually a class of reviews
  - Walkthroughs
  - Inspections
  - Round-robin reviews
  - Other small group technical assessments of the software

The Review Meeting

- Constraints
  - Between 3 and 5 people (typically) are involved
  - Advance preparation should occur, but should involve no more that 2 hours of work for each person
  - Duration should be less than two hours
- Components
  - Product - A component of software to be reviewed
  - Producer - The individual who developed the product
  - Review leader - Appointed by the project leader; evaluates the product for readiness, generates copies of product materials, and distributes them to 2 or 3 reviewers
  - Reviewers - Spend between 1 and 2 hours reviewing the product, making notes, and otherwise becoming familiar with the work
  - Recorder - The individual who records (in writing) all important issues raised during the review

Review Reporting and Recordkeeping

- Review Summary Report
  - What was reviewed?
  - Who reviewed it?
  - What were the findings and conclusions?
- Review Issues List
  - Identify the problem areas within the product
  - Serve as an action item checklist that guides the producer as corrections are made

Guidelines

- Review the product, not the producer
- Set an agenda and maintain it
- Limit debate and rebuttal
- Enunciate the problem areas, but don't attempt to solve every problem that is noted
- Take written notes
- Limit the number of participants and insist upon advance preparation
- Develop a checklist for each product that is likely to be reviewed
- Allocate resources and time schedules for FTRs
- Conduct meaningful training for all reviewers
- Review your earlier reviews (if any)

Reviewer's Preparation

- Be sure that you understand the context of the material
- Skim all product material to understand the location and the format of information
- Read the product material and annotate a hardcopy
- Pose your written comments as questions
- Avoid issues of stylea
- Inform the review leader if you cannot prepare

Results of the Review Meeting

- All attendees of the FTR must make a decision
  - Accept the product without further modification
  - Reject the product due to severe errors (and perform another review after corrections have been made)
  - Accept the product provisionally (minor corrections are needed, but no further reviews are required)
- A sign-off is completed, indicating participation and concurrence with the review team's findings